

RIPHAH INTERNATIONAL UNIVERSITY



Faculty of Computing FINAL YEAR PROJECT INITIAL PROPOSAL

GLANCE

Project Team

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Project Proposal

Project Title: GLANCE — General Look-based Assessment for Non-invasive Care & Evaluation

Background and Motivation

Many systemic diseases produce early visible signs on the body's surface, such as changes in the oral cavity, fingernails, conjunctiva, tongue, and palm, long before more invasive testing is typically ordered. In under-resourced healthcare settings, access to specialists who can recognize these signs is limited, and by the time a diagnosis is made, the disease has often progressed. GLANCE (General Look-based Assessment for Non-invasive Care & Evaluation) addresses this gap by developing a multimodal deep learning system that screens for disease indicators using only smartphone photographs of these five body surfaces.

Proposed Approach

The system uses five specialist Vision Transformer (ViT-B/16) encoders, each trained to extract features from one body surface modality. These features are combined through a custom cross-modal attention fusion transformer, built from scratch in PyTorch, which is the core research contribution of this project. The fusion model is designed to work even when one or more modalities are missing, since a real user may not always be able to photograph all five surfaces.

Explainability and Reliability

To make the system usable in practice, GLANCE incorporates Monte Carlo Dropout for uncertainty quantification, so the model can indicate when it is unsure rather than giving a falsely confident result, and GradCAM for explainability, so predictions can be visually traced back to the regions of the image that influenced them. An LLM-generated patient report translates the model's output into a plain-language summary that a non-specialist can understand and act on.

Expected Impact

By combining non-invasive image capture with an accessible smartphone interface, GLANCE aims to provide an early, low-cost screening tool that can flag potential health concerns and encourage timely follow-up with a medical professional, particularly in settings where specialist access is limited.